

Assignment No. - 1

ON

Cryptography and Network Security - SSCS 4020

TITLE: Classical Encryption Techniques

(Caesar Cipher, Monoalphabetic Cipher, Playfair Cipher, and Vigenère Cipher)

**BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY
(HONOURS.)**

SUBMITTED BY:

Name: Shah Rishabh Alpeshbhai

Enrolment: 23SS02IT181

Class: BSCIT-H – 7A - Batch 2023-27

Problem Statement – 1: Encryption and Decryption Using Caesar Cipher

Aim: To implement the Caesar Cipher algorithm for encrypting and decrypting a given plaintext using a fixed shift key.

Input:

```
def encrypt(text, key):
```

```
    result = ""
```

```
    for char in text:
```

```
        if char.isalpha():
```

```
            shift = (ord(char.upper()) - 65 + key) % 26
```

```
            result += chr(shift + 65)
```

```
        else:
```

```
            result += char
```

```
    return result
```

```
def decrypt(cipher, key):
```

```
    result = ""
```

```
    for char in cipher:
```

```
        if char.isalpha():
```

```
            shift = (ord(char.upper()) - 65 - key) % 26
```

```
            result += chr(shift + 65)
```

```
        else:
```

```
            result += char
```

```
    return result
```

```
text = input("Enter Plaintext: ")
```

```
key = int(input("Enter Key: "))
```

```
cipher = encrypt(text, key)
```

```
plain = decrypt(cipher, key)
```

```
print("Ciphertext:", cipher)
```

```
print("Decrypted Text:", plain)
```

Output:-

```
Enter Plaintext: Rishabh
Enter Key: 3
Ciphertext: ULVKDEK
Decrypted Text: RISHABH
```

Problem Statement – 2: Secure Communication Using Caesar Cipher

Aim: To simulate secure message transmission using the Caesar Cipher algorithm with user-defined shift values.

Input:

```
def encrypt(text, key):
    result = ""
    for char in text:
        if char.isalpha():
            shift = (ord(char.upper()) - 65 + key) % 26
            result += chr(shift + 65)
        else:
            result += char
    return result

def decrypt(text, key):
    result = ""
    for char in text:
        if char.isalpha():
            shift = (ord(char.upper()) - 65 - key) % 26
            result += chr(shift + 65)
        else:
            result += char
    return result

message = input("Enter Message: ")
key = int(input("Enter Key: "))
```

```
encrypted = encrypt(message, key)
decrypted = decrypt(encrypted, key)
print("Encrypted Message:", encrypted)
print("Decrypted Message:", decrypted)
```

Output:-

```
Enter Message: P P SAVANI UNIVERSITY
Enter Key: 5
Encrypted Message: U U XFAFSN ZSNAJWXNYD
Decrypted Message: P P SAVANI UNIVERSITY
```